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Review article

Microalgae-based biodiesel production and its challenges and future opportunities: A review



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ABSTRACT

The rapid depletion of fossil fuel reserves and rising greenhouse gas (GHGs) emissions have necessitated research into low-cost renewable energy sources, with a focus on algal-derived biofuels. Biodiesel, a sustainable alternative fuel, can be derived from a variety of biological sources Microalgae-based biodiesel production offers an innovative method of generating renewable and sustainable energy. By utilizing advanced cultivation techniques, such as photobioreactors and open pond systems, microalgae can be grown efficiently and in large quantities. Furthermore, various methods, such as lipid extraction and transesterification, are employed to convert the lipid-rich microalgae biomass into biodiesel. However, to make microalgae-based biodiesel production more cost-effective and sustainable, several challenges need to be addressed, such as higher production costs, reduced productivity rates, and potential environmental impacts. Future opportunities lie in creating cost-effective and sustainable methods for cultivating and harvesting microalgae, improving lipid yields, and creating effective conversion methods. Overall, microalgae-based biodiesel production shows promise as a renewable and sustainable energy source. However, further research and development must take place to realize its full potential and reap its rewards.

1. Introduction

Global warming, resulting from greenhouse gas (GHGs) emissions, is a significant environmental challenge in the twenty-first century. Microalgae-based biodiesel production offers an innovative renewable energy solution utilizing small, single-celled organisms known as microalgae. These microalgae transform sunlight, carbon dioxide, and nutrients into organic compounds such as lipids that can then be used to make biodiesel [1]. Microalgae hold numerous advantages over other biodiesel sources, such as traditional oil crops like wheat, palm, corn, soybeans, sugarcane, rapeseed, and maize. They can thrive in various environments, including saltwater, freshwater, and wastewater, exhibiting high biomass production rates [2–4]. One of their key benefits is that they can be cultivated on non-arable land without competing with food crops for resources. Production of microalgae-based biodiesel requires several steps, and this study aims to provide a comprehensive understanding of the process. The step-by-step framework includes cultivation, harvesting, microalgae cell disruption, lipid extraction, transesterification, and the challenges and future opportunities associated with microalgae biodiesel production [5]. The cultivation stage focuses on providing optimal growth conditions for microalgae, including light, temperature, nutrients, and carbon dioxide [6–8]. After cultivation, harvesting methods encompass various techniques, such as centrifugation, filtration, and flocculation, to extract microalgae from their growth medium [9–12]. Following harvesting, microalgae cell disruption involves breaking open microalgae cells to gain access to their intracellular components, including lipids, proteins, and carbohydrates [13–15]. There are various methods for disrupting microalgae cells: mechanical, chemical, and biological methods [16,17]. The next step, lipid extraction from microalgae, involves extracting oils or fats from microalgae cells for further processing [18,19]. Microalgal oil extracted is then converted to biodiesel through transesterification, which is an essential process used in biodiesel production. It involves the conversion of triglycerides derived from plant or animal fats to fatty acid methyl esters (FAMEs), the main constituents of biodiesel [20–22]. Microalgae-based biodiesel production holds great promise as an environmentally sustainable and climate-friendly alternative to fossil fuels, as it reduces GHGs and alleviates the burden on non-renewable resources [23–26]. However, realizing its full potential as an economically and environmentally viable energy source requires overcoming several challenges [27,28].

The major hurdle is the high cost associated with cultivating, harvesting, and extracting microalgae oil for biodiesel production. To compete with fossil fuels, efforts should prioritize cost reduction and efficiency improvement in microalgae biodiesel production [27,28]. An additional challenge lies in establishing large-scale production facilities to meet market demands for microalgae-based biodiesel [29]. While

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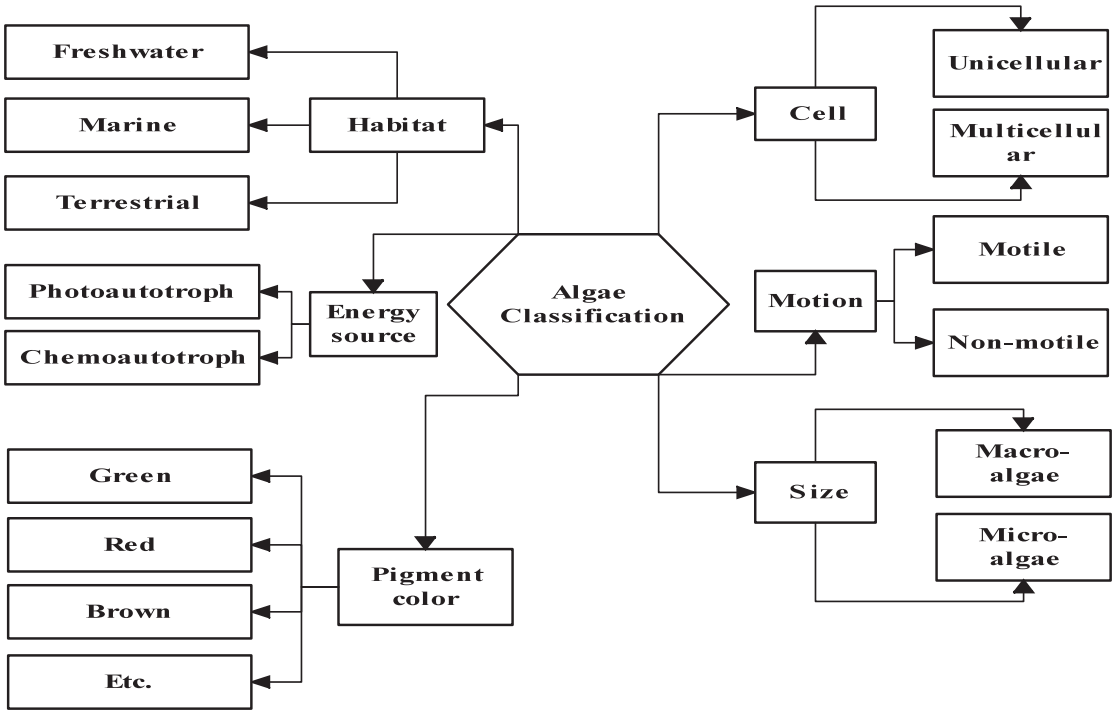


Fig. 1. Taxonomic classification of algae.

microalgae can be grown on a small scale, their limited quantities of biodiesel production make large-scale facilities crucial [30,31]. Moreover, ensuring the sustainability of microalgae biodiesel production raises concerns related to water usage and nutrient supply during cultivation [32–35]. However, to make microalgae-based biodiesel production more cost-effective and sustainable, several challenges need to be addressed, such as higher production costs, reduced productivity rates, and potential environmental impacts. Nonetheless, the future of microalgae-based biodiesel remains promising [36–38]. Addressing challenges in production efficiency, scaling up, and sustainable practices can unlock the full potential of microalgae biodiesel. Strategies like developing resilient microalgae strains capable of thriving in diverse conditions and utilizing waste materials like wastewater and carbon dioxide for cultivation could significantly contribute to cost reduction and environmental improvement [39–41].

In summary, this study aims to provide a comprehensive understanding of microalgae-based biodiesel production, from cultivation to transesterification. By addressing the challenges and harnessing the advantages of microalgae, this renewable energy solution can play a vital role in combating climate change and reducing our dependence on non-renewable resources.

2. An overview of algae

Algae are a highly diverse group of photosynthetic organisms found in a variety of habitats. Algal blooms can occur in freshwater, saltwater, sediment, and even the atmosphere [8,42]. All types of microalgae, from single-celled microalgae to large multicellular seaweeds can be found on the earth's outermost layer [43–45]. Algae are one of the primary producers in aquatic ecosystems such as lakes, rivers, and oceans. Through photosynthesis, algae use energy from the sun and nutrients from water to convert carbon dioxide into organic matter. Organic matter is the basis of the food chain, providing nutrition and energy to organisms like fish, crustaceans, and zooplankton. Algae are not only essential primary producers but also major oxygen producers. Aquatic organisms require oxygen to survive, which algae release through photosynthesis. Experts estimate that up to 70% of Earth's oxygen comes from algae and other marine plants [46,47].

2.1. Classification of algae

Algae can be classified according to various characteristics such as their habitat, energy source, pigment color, size, cell type, and movement, as shown in Fig. 1. The methods of classification are discussed below.

- Habitat: Algae can be divided into different classes depending on where they live, such as freshwater, marine, or terrestrial environments [48–50].
- Energy Source: Algae can be classified based on their source of energy. Some algae are photosynthetic, utilizing sunlight as their energy source, while others are chemosynthetic, deriving energy from chemical reactions. Additionally, there are mixotrophic algae, which use a combination of photosynthesis and external organic carbon for energy [51,52].
- Pigment Color: Algae can be divided based on their pigment color, which gives them their unique appearance. Examples of pigment colors are green, brown, red, and blue–green [53–55].
- Size: Algae are often classified based on their size. These range from single-celled microalgae to seaweeds that may reach several metres long [56–58].
- Cell Type: Algae can be divided into two major classifications based on their cell type: prokaryotic (unicellular) or eukaryotic (multicellular). The structure of prokaryotic algae is simpler, whereas that of eukaryotic algae is more complex and larger [59–61].
- Motion: Algae can be divided into two main categories, depending on their mobility: motile or non-motile. Motile algae have flagella to propel themselves through water currents, while non-motile varieties remain immobile on a substrate [62–64].

Table 1
Key features of microalgae species.

Feature	Description	Reference
Size	Microalgae typically range in size between 2–50 μm.	Elisabeth et al. [65]
Photosynthesis	Microalgae are photosynthetic organisms that use sunlight, carbon dioxide, and water as energy sources to produce organic compounds.	Goswami et al. [7], Serrà et al. [66] and Touliabah et al. [67]
Production rate	Microalgae have an extraordinary growth rate and can double their biomass within 24 h under optimal conditions.	Leong et al. [68] and Qari et al. [69]
Nutritional content	Microalgae are packed with vital nutrients such as proteins, lipids, vitamins, and minerals.	Lucakova et al. [70] and Wu et al. [71]
Biodiversity	Microalgae are a diverse group of organisms with over 70,000 known species.	Neofotis et al. [72] and Vieira et al. [73]
Habitat	Microalgae can be found in different aquatic environments, including freshwater, marine, and brackish waters.	Enamala et al. [74], Manganelli [75] and Sendra et al. [76]
Bio-active compounds	Microalgae can produce bioactive compounds with numerous applications across medicine, cosmetics, and other industries.	Sonkamble and Wagh [77]
Eco benefits	Microalgae can be utilized for carbon sequestration and wastewater treatment applications, among other environmental benefits.	Ali et al. [78] and Orejuela-Escobar et al. [79]

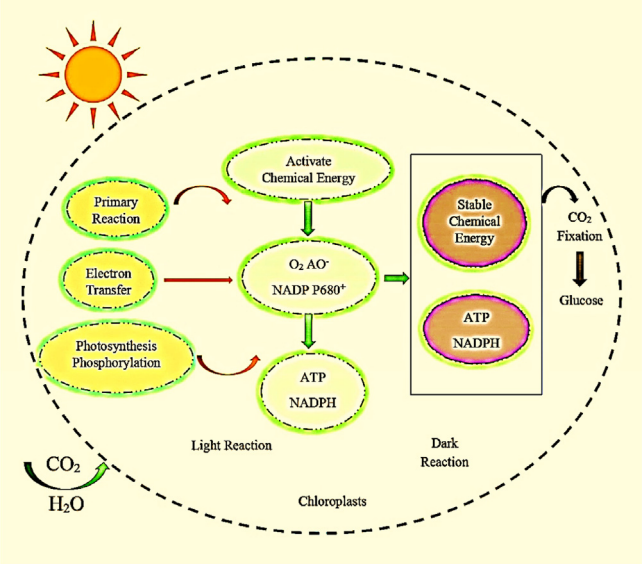


Fig. 2. Process of photosynthesis in algae.

2.2. Features of microalgae

Microalgae are organisms found in aquatic environments like oceans, lakes, and ponds. Each variety exhibits unique features, as shown in Table 1.

2.3. The process of photosynthesis in algae

Algae are photosynthetic organisms that derive the majority of their energy from sunlight. In photosynthesis, light energy is converted into chemical energy in the form of glucose, which microalgae use for growth and metabolic processes [8,80,81]. In the course of photosynthesis, algae undergo two distinct reactions: the light reaction and the dark reaction [82]. Light energy is used by algae to generate chemical energy in the form of ATP and NADPH molecules. Light reactions also involve the activation of P680 molecules that absorb light energy [83,84]. In addition, under the dark reaction or Calvin cycle, microalgae use the energy produced by photosynthesis to fix carbon dioxide (CO_2) into glucose for production. This process requires both ATP and NADPH as energy sources for a series of chemical reactions [85–87]. In addition, photosynthesis in algae not only fixes carbon dioxide and produces glucose, but it also releases oxygen (O_2) and converts NADP+ to NADPH, as shown in Fig. 2. All of these reactions occur within the chloroplasts of microalgae, which contain pigments, enzymes, and other molecules necessary for photosynthesis [88].

2.4. Nutritional composition of algae

Algae are aquatic photosynthetic organisms with diverse species and growth conditions that vary considerably in their nutritional composition, with amounts of proteins, carbohydrates, and lipids typically differing based on species and growth conditions, as shown in Fig. 3.

2.5. Advantages and disadvantages of microalgae as a resource

Microalgae are an increasingly popular source of food, feed, and biofuel. There are many advantages to using microalgae as a resource, but there are also drawbacks, as shown in Table 2.

In conclusion, the overall findings emphasize the significance of algae as diverse photosynthetic organisms thriving in various habitats. Their vital role as primary producers in aquatic ecosystems, releasing oxygen and providing essential nutrition, cannot be understated. The classification of algae based on habitat, energy source, size, and other factors highlights their versatility and adaptability. Microalgae, with their nutrient-rich composition and unique attributes, present considerable potential. Through the process of photosynthesis, algae efficiently convert light energy into chemical energy, yielding glucose and oxygen. Despite facing challenges in commercial production, the advantages of high productivity and potential sustainability as a resource for food, feed, and biofuel make microalgae an area of promising research and technological development.

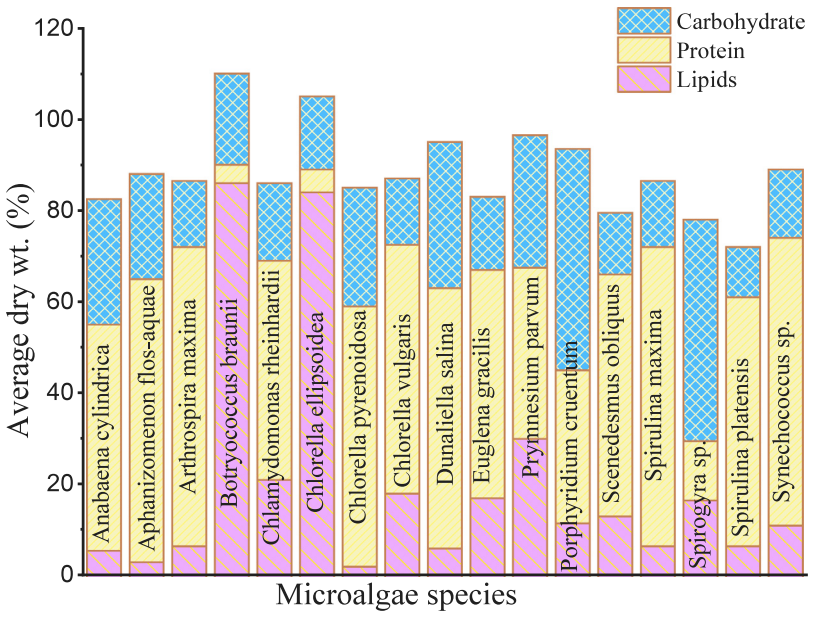


Fig. 3. Variation in protein, carbohydrate, and lipid contents among different algae species [89].

Table 2
Advantages and disadvantages of using microalgae for biodiesel production [12,74].

Advantage	Disadvantage
• High productivity • Carbon neutral • Can be grown in various environments • Implement an efficient nutrient cycling cycle • The significant decline in the competitiveness of food compared to fuel • Produces biofuels and other valuable products • Reduce greenhouse gas emissions • Use in wastewater treatment applications • Aquaculture nutrition can be provided as a source of nutrition for aquatic farms • Consumption of algae by pregnant women may prove advantageous due to their rich iron (Fe) content • Cyanobacteria are capable of fixing nitrogen from the atmosphere to form fertile soil for plants to grow • Solar conversion efficiency is 10 times greater compared to plants • Premium-grade lipid composition may help promote overall well-being • Open ponds can be used to cultivate • A system covering several kilometers	• High capital cost • Harvesting and drying costs • Contamination and stability issues • Scaling up for commercial production can be challenging • Limited market and strong competition from other feedstocks • Land and water usage requirements • Expertise and infrastructure are needed for success • High temperatures pose a severe threat to open-pond farming operations

3. Algae biomass cultivation

Algae cultivation refers to the practice of growing algae for various uses such as food, fuel, medicine, and waste disposal. Algae are photosynthetic organisms with rapid growth rates and potential high productivity, making them an attractive option for sustainable and scalable production processes [90]. Algal cultivation can be accomplished using various methods, such as open ponds, closed photobioreactors (PBRs), and hybrid systems [91], as shown in Fig. 4. Using an open pond is the simplest and cheapest method, however it can also become polluted and affected by changes in the environment. PBRs offer greater control over pollution and its effect on the environment; however, their operating and maintenance expenses tend to be higher. Algae cultivation requires an appropriate growth medium that contains nutrients like nitrogen and phosphorus as well as trace elements like iron and zinc. The type of medium chosen depends on both the species being grown and its desired use [92]. Growing algae has numerous potential advantages, including reduced greenhouse gas emissions, improved water quality, increased food production potential, and a positive impact on the environment by mitigating nutrient pollution and sequestering carbon dioxide. Algae farming presents its own set of unique challenges, including increasing production while mitigating environmental effects [93].

3.1. Phototrophic cultivation

Phototrophic cultivation (photoautotrophic farming) is an innovative technique for growing algae using light as its energy source, employing photoautotrophic organisms like algae and cyanobacteria that convert light energy to chemical energy through photosynthesis. Algae can be cultivated in diverse environments, including open ponds, PBRs, and open ocean environments. An appropriate cultivation system selection depends on several factor, including desired algae species, available resources, and economic feasibility [94]. Phototrophic cultivation involves cultivating algae in an enclosed system with optimal lighting and nutrient conditions, designed to capture and reuse carbon dioxide or wastewater as needed. By using such systems, waste may be minimized and nutrient reused to maximize efficiency and minimize waste [95]. Phototrophic algae growth success relies on virous factors, including light intensity, temperature, availability of nutrients, and pH level. Growth and productivity can be maximized with careful monitoring and management [96].

3.1.1. Natural cultivation (open pond)

Open-pond algae cultivation refers to the process of cultivating aquatic organisms in an open-air pond made of concrete or lined with

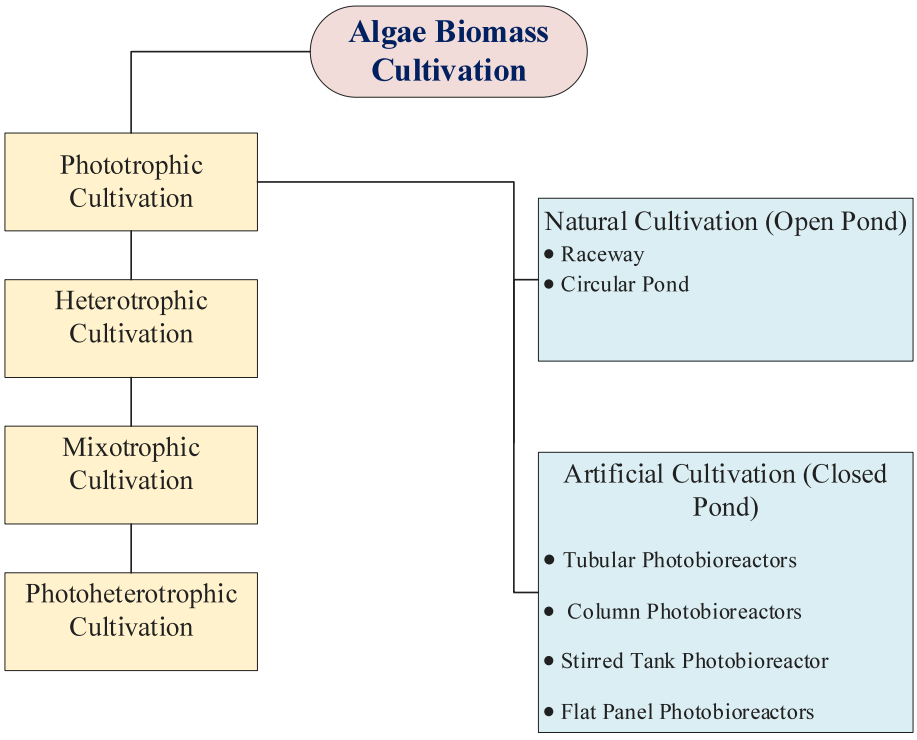


Fig. 4. Schematic representation of the cultivation process for algal biomass production.

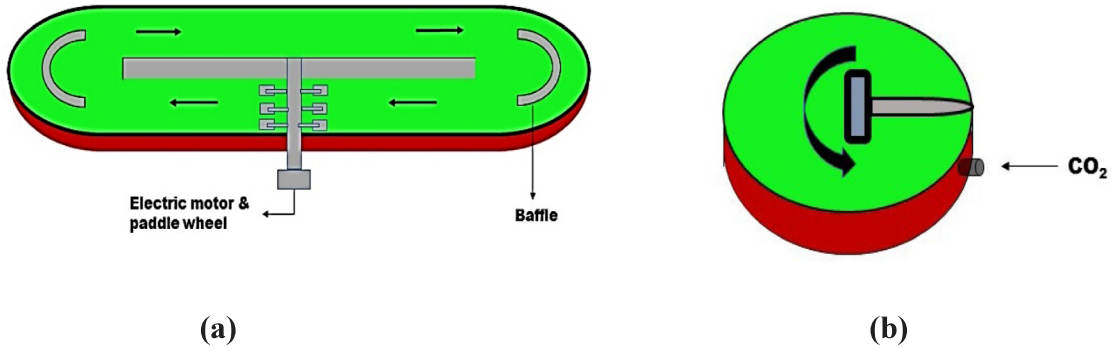


Fig. 5. (a) Raceway in the open pond system (b) Circular Pond in the open pond system.

synthetic materials. It is widely employed and relatively straightforward [96]. Initial operations involve filling a pond with water and adding an algae culture [97]. Once planted, algae are given ample sunlight and additional nutrients such as nitrogen or phosphorus to promote their growth. Fig. 5(a) and (b) shown, raceways, and open circular ponds respectively. Even though cultivating algae in open ponds can be cost-effective, certain drawbacks must be considered when considering this method of cultivation. One of the main challenges associated with open pond cultivation is contamination from external sources, which could potentially lower both the quality and quantity of production [98]. Furthermore, size constraints and the limited availability of nutrients may hamper production in a pond [99].

3.1.2. Artificial cultivation (closed pond or PBRs)

Closed-pond algae cultivation or photobioreactors cultivation involves cultivating algae within an enclosed system. PBRs cultivation offers several distinct advantages over open-pond farming, including greater environmental control, higher yields, and decreased contamination risks [100]. Algae are typically grown within an enclosed container where temperature, light intensity, and nutrients can all be closely controlled for maximum productivity [101]. Containers used for cultivating algae often consist of glass or plastic materials and may be cylindrical or tubular [102]. Lighting sources include natural sunlight or artificial lighting sources [78]. Algae are typically grown using nutrient-rich media. Closed pond systems are also less prone to contamination from outside sources, such as dust, insects, and other pollutants [103] Unfortunately, this method has its own set of drawbacks, such as the higher capital and operating costs associated with developing and maintaining closed systems. The advantages and disadvantages of both open and closed systems, as shown in Table 3.

3.1.2.1. Tubular photobioreactors (T-PBRs). T-PBRs are closed-system algae cultivation technologies comprised of interconnecting transparent tubes through which algae-filled water circulates [107]. Plastic or glass tubes arranged within an enclosure typically serve as the ideal environment for growing dense biomass, as shown in Fig. 6. The tubes have been specifically designed to ensure maximum light distribution with minimal shading, and the water flow rate can easily be adjusted to achieve optimal nutrient uptake and growth. Bioreactors offer many advantages for growing algae. Scalability is another added advantage. To increase production, multiple tubes may be linked in parallel and connected [108]. T-BPRs provide simple and energy-efficient methods of growing algae when compared to other cultivation techniques. While

Table 3
Comparison of advantages and disadvantages between open and closed pond systems [104–106].

Pond system	Advantages	Disadvantages
Open pond	• Low cost and easy to construct • Natural sunlight is used for algae growth • Large surface area for algae growth • Simple maintenance requirements • Economically beneficial • More durable • Higher production volume possible • Simple scaling • Effective exchange of gases with the atmosphere, including the release of oxygen	• Exposed to contamination from outside organisms • High rates of water evaporation • Limited control over environmental conditions • Low cell densities and productivity • Limited to hardy species • Fixation rates remain low
Closed pond	• Improved environmental conditions • Reduced risk of contamination • Increased algae productivity • Reduce water evaporation rates • Compatible for use in various climate conditions • High mixing (vertical tube PBRs) • Higher cell densities and productivities • Initial investment costs are low (Considering the short term)	• Higher capital and operational expenses • Growth requires artificial lighting sources • Maintaining pH or oxygen gradients (horizontal tube PBRs) can be challenging • complex cleaning processes • Scaling up can be difficult • Shear forces of significant magnitude • Biofilm formation on inner surfaces

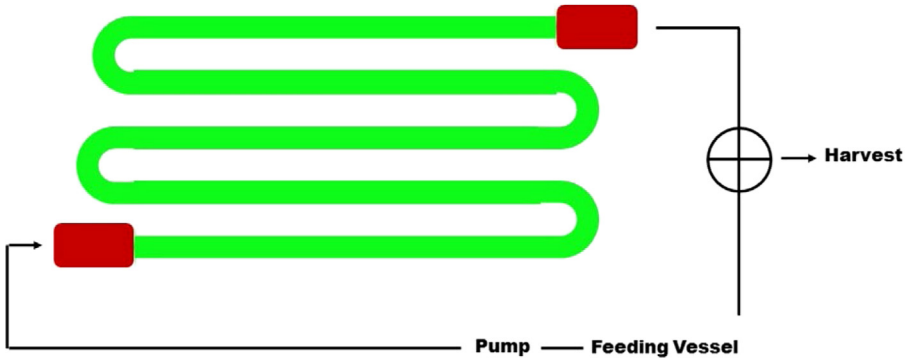


Fig. 6. Tubular photobioreactors for microalgae cultivation.

they have some drawbacks, including the need for regular cleaning to eliminate biofilm buildup and the potential for higher material costs for their construction and associated infrastructure [109].

3.1.2.2. Column photobioreactors (C-PBRs). C-PBRs are enclosed systems used for cultivating algae. It consists of an opaque tube with an enclosed top that houses an inner chamber wherein algae grow under controlled conditions [110], as shown in Fig. 7. Bioreactors provide multiple advantages to farmers when it comes to producing biomass yields. The main advantage of these systems is that they provide effective light penetration and gas transfer, which produces a higher biomass yield [111]. C-PBRs offer several advantages over open ones, including reduced contamination risks and greater control over culture parameters like light intensity, temperature, and nutrient availability. However, some drawbacks are associated with bioreactors, including high capital and operating costs as well as light gradients that could result in uneven growth within their columns [31].

3.1.2.3. Stirred tank photobioreactor (ST-PBRs). ST-PBRs are specifically designed to grow microalgae, cyanobacteria, and other photosynthetic microorganisms [107,112]. Bioreactors consist of cylindrical vessels fitted with impellers powered by motorized electric current to mix culture medium while providing enough light exposure, enabling microorganisms to flourish for photosynthesis, as shown in Fig. 8. Photobioreactors feature light sources that provide enough intensity and spectrum distribution to support photosynthesis [113,114]. The culture medium is continuously or intermittently circulated through this bioreactor to supply essential nutrients while eliminating waste products. To maximize productivity, pH, temperature, and dissolved oxygen are adjusted. ST-PBRs offer many advantages over other types of photobioreactors, including easy scalability, increased biomass productivity, and efficient gas transfer [115]. Unfortunately, they also

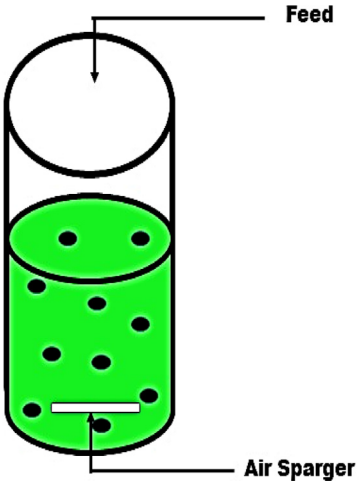


Fig. 7. Column photobioreactors for microalgae cultivation.

pose certain drawbacks that need to be considered, including higher energy usage, reduced light attenuation, and mechanical damage risk to microorganisms [116].

3.1.2.4. Flat Panel Photobioreactors (FP-PBRs). FP-PBRs are closed cultivation systems consisting of an open container equipped with plastic or glass panels arranged vertically to allow sunlight access [108,117], as shown in Fig. 9. bioreactors are designed to optimize lighting conditions and CO_2 availability to promote efficient growth of microorganisms such as algae or cyanobacteria suspended in nutrient-rich medium,

Table 4
Comparison of various types of Photobioreactor for microalgae cultivation [108,121].

Photobioreactor type	Advantages	Disadvantages
Tubular	• Good light distribution • Easy to scale up • High biomass productivity	• High energy consumption for pumping fluids • Prone to clogging • Limited temperature control
Column	• Good mixing and aeration • High biomass productivity • Easy to control temperature and pH	• Limited light penetration • High shear stress on cells • Difficult to scale up
Stirred tank	• Good mixing and aeration • Easy to control temperature and pH • Can handle a variety of cultures	• Limited light penetration • High energy consumption • Low biomass productivity
Flat panel	• High light penetration • Low energy consumption • Easy to control temperature and pH	• Limited biomass productivity • Prone to biofilm formation • Difficult to scale up

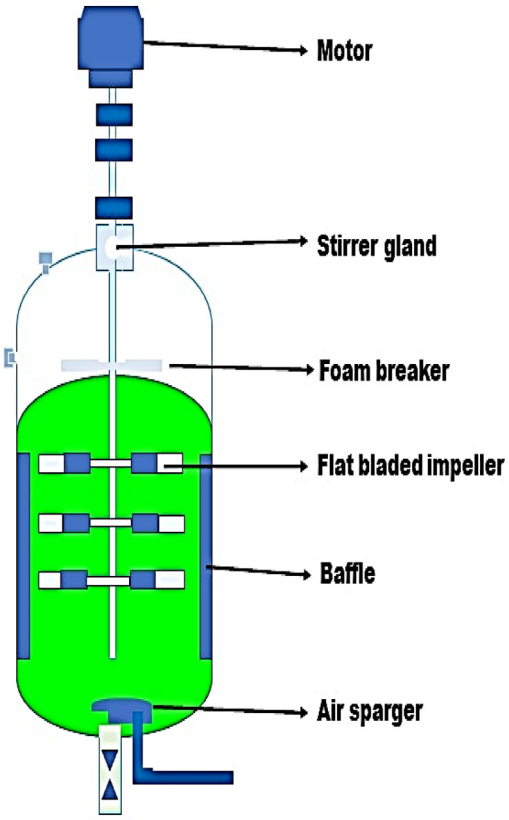


Fig. 8. Stirred tank photobioreactor for microalgae cultivation.

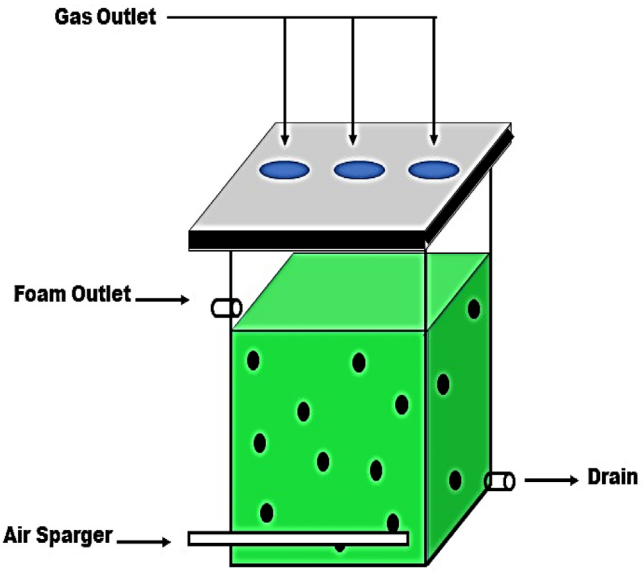


Fig. 9. Flat Panel Photobioreactors for microalgae cultivation.

with bioreactors aeration continuously maintaining optimal oxygen levels for harvesting purposes. Harvesting occurs through regular drainage of the medium and harvesting by centrifugation or filtration, depending on its suitability. FP-PBRs offer numerous advantages over open pond systems, including greater biomass production yields, improved control over growth conditions, decreased water usage, and urban applications such as rooftop installations due to their compact designs [118,119]. However, their higher capital costs and maintenance requirements limit widespread adoption [102,114,120].

3.1.2.5. Advantages and disadvantages of different types of photobioreactors. Photobioreactors are devices that use light to cultivate microorganisms such as algae for biofuel production or wastewater treatment purposes. The advantages and disadvantages of different types of photobioreactors, as shown in Table 4.

3.2. Heterotrophic cultivation

Heterotrophic cultivation involves growing algae with carbon sources other than photosynthesis, such as sugars and acids [122,123]. This method can be especially effective in environments in which algae does not receive sufficient light or carbon dioxide, such as indoor spaces and shaded outdoor areas. Heterotrophic cultivation can result in higher yields and faster growth rates compared to autotrophic cultivation [100,124]. However, this approach could prove more expensive as it requires constant access to organic carbon sources. Heterotrophic cultivation may be combined with other cultivation methods, such as photoautotrophic or mixotrophic cultivation, to optimize growth conditions for each type of algae [125,126].

3.3. Mixotrophic cultivation

Mixotrophic cultivation involves growing microorganisms using both organic and inorganic nutrients to support their growth, like algae, bacteria, and fungi [127,128]. The process utilizes carbon dioxide and organic compounds as energy sources. The advantages of mixotrophic cultivation over other cultivation methods include higher growth rates, more productivity, reduced waste production, and more efficient resource use [8,124,129]. Mixotrophic cultivation requires careful oversight of its nutrient levels, as the ratio between organic and inorganic nutrients can have an impactful impact on both the growth rate and productivity of microorganisms [40,130].

Table 5
Comparative analysis of various cultivation methods for algae [105,135,136].

Cultivation method	Advantage	Disadvantage
Phototrophic cultivation	• Do not require expensive carbon sources • Eco friendly	• It requires specialized equipment to harvest light • Not all microorganisms are photosynthetic
Heterotrophic cultivation	• Grow a wide array of microorganisms • Light or specialized equipment is not required	• Expensive carbon sources may be required. • produces waste that needs to be disposed of properly
Mixotrophic cultivation	• Utilize both light and organic carbon sources for flexibility in nutrient sourcing • Enhancing growth rates when compared with phototrophic or heterotrophic cultivation alone	• Requires specialized equipment for light harvesting • Nutrient requirements can be more complex • More challenging to optimize
Photoheterotrophic cultivation	• Utilizes organic carbon sources while using light as an energy source • Increased growth rates when compared with heterotrophic cultivation alone	• Requires specialized equipment for light harvesting • More complex nutrient requirements can make it more challenging to optimize

3.4. Photoheterotrophic cultivation

A photoheterotrophic cultivation involves organisms utilizing both light energy and organic compounds for growth. Abreu et al. [131] and Patel et al. [132]. Photoheterotrophic cultivation uses light energy and organic compounds as carbon and energy sources, providing photoheterotrophic microorganisms like purple and green bacteria with an optimal environment. Typically, it is used when cultivating organisms capable of photoheterotrophic metabolism, such as phototrophs. Photoheterotrophic cultivation growth media often contain organic substances like glucose or sucrose to provide energy sources to the microorganisms being grown [133,134]. Photoheterotrophic cultivation requires not only organic compounds but also light sources like fluorescent or incandescent lamps for optimal results. Microorganisms are cultured at different intensities and durations depending on the light used. Photoheterotrophic cultivation offers several advantages, such as high growth rates at reduced costs, sustainability, flexibility, and potential bioremediation solutions [120].

3.5. Advantages and disadvantages of different types of algal cultivation methods

Advantages and disadvantages of various algal cultivation methods depend on factors like cost, scalability, and productivity, as shown in Table 5.

3.6. Comparison of open pond and closed pond systems

Algae cultivation can be done in two primary ways: in open and closed pond systems. Both approaches are necessary to grow algae. Table 6 details these basic processes involved with phototrophic cultivation.

3.7. General condition for microalgae growth

Microalgae growth requires specific conditions to be optimal, but following general guidelines can provide some insight into the optimal conditions.

- Light Intensity: Light intensity and duration have an impactful impact on growth rates. In general, an exposure intensity between 2000–4000 lux is recommended [141,142].
- Light Duration: Microalgae require light for photosynthesis. Microalgae need the appropriate light duration to maximize their photosynthetic efficiency, growth, and productivity. Too little or too much exposure can negatively affect their growth and development. Typically, they grow best under constant light/dark cycles of 12:12 or 16:8 h [143].
- Temperature: The ideal temperature for microalgae cultivation varies with species. Most strains survive at temperatures between 20–30 °C [144,145].
- Nutrients: Microalgae require essential nutrients like nitrogen, phosphorus, and potassium for their growth. While ideal concentrations may differ depending on the species of microalga being grown, As a basic guideline, attempt to maintain the levels of nitrogen and phosphorus between 50–150 mg/L and 5–20 mg/L, respectively [146,147].
- pH: The ideal pH range for growing microalgae depends on its species. Typically, most organisms survive within the pH range of 7–9 [148,149].
- CO_2 Concentration: Microalgae use CO2 as a carbon source during photosynthesis, and higher CO_2 concentrations can enhance growth rates. while optimal CO_2 concentration levels vary depending on species. A range of 1%–2% for CO_2 would serve as a standard guide [150].
- Salinity: Salinity requirements for microalgae cultivation depend on its species and growth conditions. some common ranges for cultivation include
  - Freshwater microalgae typically range from 0–0.5 ppm [151]
  - Brackish water microalgae have salinity ranges of 0.5–30 ppt [152,153].
  - Marine microalgae have ranges of 30–40 ppt [154]

Optimum Condition for microalgae growth, as shown in Table 7

In conclusion, the overall findings emphasize the diverse and promising aspects of algae cultivation. The phototrophic approach, whether in open ponds or closed PBRs, offers great potential for sustainable production. Different closed PBRs like T-PBRs and C-PBRs provide effective means for controlled growth. Additionally, alternative methods like heterotrophic, mixotrophic, and photoheterotrophic cultivation expand the scope of algae utilization. Understanding the critical factors influencing microalgae growth, such as light, temperature, nutrients, pH, CO2, and salinity, is essential for successful cultivation. By choosing the right method and optimizing conditions, algae cultivation can revolutionize various industries with its eco-friendly and efficient solutions.

4. Microalgae harvesting

Microalgae harvesting is the method of removing microalgae cells from their culture medium and using them to make things like biofuels, food supplements, and medicines [103,155]. Harvesting procedures for microalgae production are typically the most energy-intensive and expensive, which has a significant impact on the overall economics of production systems [156,157]. Consequently, it is crucial to create efficient and cost-effective techniques for harvesting microalgae. Microalgae harvesting methods include physical, chemical, biological, and magnetic methods, as shown in Fig. 10.

Table 6
Comparison of steps involved in open pond and closed pond systems for algae cultivation [137–140].

Process step	Open pond system	Closed pond system
Site selection	Flat and sunny area with access to water and a gentle slope to assist drainage	The site can be chosen based on the availability of water but must be enclosed and equipped with aeration and mixing systems
Construction	Excavate the shallow pond with sloped walls and install the liner	Construct a pond using concrete or other materials, with airtight walls and bottom
Filling	Fill the pond with water and add nutrients	Fill the pond with water and add nutrients
Algae strain selection	Select a suitable algae strain for the environment and target the product	Select a suitable algae strain for the environment and target the product
Inoculation	Inoculate the algae strain into the pond	Inoculate the algae strain into the bioreactor
Nutrient addition	Add nutrients (fertilizers) to the water	Add nutrients (fertilizers) to the water
pH adjustment	Adjust the pH to optimize growth	Adjust the pH to optimize growth
Light exposure	Ensure adequate sunlight exposure	Provide artificial light or optimize natural light exposure
Mixing	Natural aeration, mechanical mixing, or paddle wheels	Mechanical mixing (stirrers or pumps)
Maintenance	Monitor and maintain the pond's water quality, temperature, and other parameters	Monitor and maintain the bioreactor's temperature, pH, and other parameters
Harvesting	Skim the surface of the pond to collect algae or drain the pond and collect biomass from bottom	Use mechanical or centrifugal methods to separate algae from the water, then dry and package biomass
Dewatering	Dewater the harvested algae biomass to remove excess water	Dewater the harvested algae biomass to remove excess water
Processing	Process the algae biomass into the desired product such as biofuels, animal feed, or cosmetics	Process the algae biomass into the desired product such as biofuels, animal feed or cosmetics
Reseeding	Re-seed the pond with fresh algae to start a new growth cycle	Re-seed the bioreactor with fresh algae to start a new growth cycle

Table 7
Optimum conditions for cultivating algae.

Factor	Ideal condition
Temperature	20–30 °C
Light intensity	2000–4000 lux
Light duration	12:12 or 16:8 h (light-dark cycles)
CO_2 concentration	2%–10%
Nutrients	Nitrogen, phosphorus, potassium, and micronutrients
pH	7.0–9.0
Salinity	Varies with microalgae species; typically, between 10–30 g/L

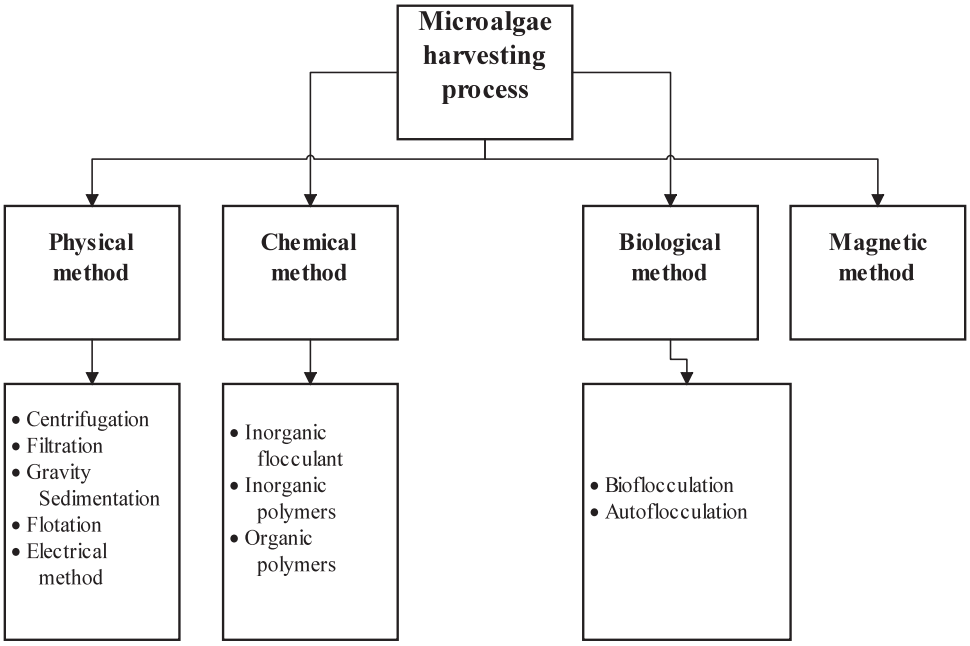


Fig. 10. Flow chart illustrating the classification process for microalgae harvesting.

4.1. The main technologies for microalgae harvesting

There are various technologies for harvesting microalgae, including physical, chemical, biological, and magnetic methods [158]. Each method has advantages and disadvantages. The selection of an efficient harvesting method is based on microalgae species, culture conditions, and intended uses.

4.1.1. Physical method

4.1.1.1. Centrifugation. Centrifugation is an efficient and common technique for harvesting microalgae that uses centrifugal force to separate the microalgae from their growth medium [159]. The method involves rapidly spinning a container containing microalgae until their heavier cells sink to the bottom, pushing any lighter growth media molecules upward [160,161]. Microalgae can then be harvested directly from this collection point. The efficiency of the centrifugation method of harvesting is significantly influenced by the rotational speed of the centrifuge, the duration of the rotational process, and the size and density of the microalgae cells. High separation efficiency can be achieved by optimizing these parameters. Centrifugation is an extremely efficient microalgae harvesting method capable of quickly reaching high biomass recovery rates in a short amount of time. Unfortunately, centrifugation is energy-intensive and costly to run in larger-scale operations compared to alternative techniques like flocculation or filtration [10,162]. Despite this, it is still a crucial instrument for small to medium-scale harvesting applications.

4.1.1.2. Filtration. Filtration is an effective and widely used technology for harvesting microalgae in large-scale production systems. A porous material, such as a filter or membrane, is used in the filtration technique to separate solids (microalgae cells) from a liquid (culture media). The size of the filter membrane or pore depends on the biomass's particle size. Some examples of filters used for filtration are sand filters, mesh filters, and ceramic filters. The process involves filtering microalgal cultures to separate the microalgae from the liquid medium [163]. Filtration processes may involve the use of various filters, including microfiltration or ultrafiltration membranes, depending on the size and purity level desired for microalgae harvest. Filtration has its limitations, such as filter clogging, high energy usage, and the need for regular maintenance [164]. However, there are some limitations that should be taken into consideration. Filtration can be time consuming and required periodic cleaning or replacement of filters due to clogging and fouling. The presence of small particles or debris in the culture media also affects the effectiveness of filtration process. The desired result can be achieved by monitoring and optimizing the filtration process. These limitations can be overcome by combining them with other techniques like sedimentation, flocculation, and centrifugation to increase microalgae harvesting efficiency [160,165].

4.1.1.3. Gravity sedimentation. Gravity sedimentation process involves allowing the microalgae suspension to stand undisturbed in a settling tank or a large container. As the time passes, the microalgae cell settles to the bottom of tank due to higher density. Microalgae may be easily and affordably harvested using gravity sedimentation. This technique involves allowing the culture to settle gradually under gravity before collecting its biomass at the bottom [166,167]. Its effectiveness depends on factors like microalgae size, density, the concentration of the culture medium used, viscosity of growth medium and the duration of sedimentation. However, small or lightweight microalgae often remain suspended in their liquid medium and become difficult to separate from it. Performance of gravity sedimentation can be enhanced by employing various technique. Coagulation and flocculation agents can be added to promote the formation of larger aggregates, which settle faster. The use of chelating agent and pH adjustment also help in flocculation. Furthermore, the use of inclined settler or clarifier increases the sedimentation area and helps in improving separation efficiency. Therefore, this technique should generally be employed for denser microalgae [168].

4.1.1.4. Flotation. Microalgae can be extracted from their growth medium using the flotation process. In this method, gas bubbles like air or nitrogen are injected into the microalgae's growing medium, causing the cells to rise to the surface [169,170]. Each microalgae cell is buoyed upwards towards the surface of its medium by the gas bubbles that attach to it. Once microalgae cells reach the surface, they can be collected using a scraper or skimmer. Factors that influence the efficiency of microalgae harvesting of flotation is the selections of flotations agents. Commonly used flotation agents include surfactants, polymers, or organic compound that can generate stable air bubble and facilitate the attachment of bubbles to microalgae cells. Flotation harvesting techniques tend to be preferred over other techniques since they are fast, other factors such as pH level, temperature, mixing intensity, and concentration of microalgae in culture medium also affect the flotation efficiency. To achieve high harvesting efficiency, optimal condition for these parameters for each specific microalgae species need to be determined. Flotation harvesting techniques tend to be preferred over other techniques since they are fast, simple, and cost-effective [171,172]. Furthermore, flotation requires less energy compared with centrifugation and filtration methods, making it an attractive option for large-scale microalgae cultivation [165]. However, its success may depend on multiple factors, including cell size, density, concentration in the growth medium, and the type of gas used during flotation [120,158,173].

4.1.1.5. Electrical methods. The electrical method of microalgae harvesting uses an electric field to separate microalgae from their growth medium [158]. This method takes advantage of the fact that growing media is neutrally charged, whereas microalgae cells have slightly negative charges on their surfaces [174,175]. Microalgae cells are induced by an electric field to move towards an electrode that is negatively charged, where they can be removed from their growing medium. There are several types electrical method such as Electrophoresis is commonly used electrical method, which utilizes the principle of charge differences between microalgae cells and the surrounding medium. As the electric field is applied, the charged microalgae cells migrate towards the oppositely charged electrode, allowing for their selective separation and concentration. Such method is effective for harvesting small-sized microalgae. Electroflocculation is another electrical method in which an electric field induces the aggregation of microalgae cells into larger flocs or clusters. This process is facilitated by the addition of chemical coagulants or flocculants, which helps in the bridging bond between microalgae cells and form flocs. This formed floc can be easily separated from the culture medium through sedimentation or flotation. Electrocoagulation is another electrical method, which that utilizes electric fields to destabilize the microalgae cells and promote their aggregation. In this method, metal electrodes are used to release metal ions into the culture medium, causing destabilization and subsequent coagulation of microalgae cells. The resulting coagulated mass can be separated using conventional methods such as sedimentation or filtration. The efficiency of this method depends on several factors, such as the strength, concentration, and composition of microalgae cells as well as the characteristics of their growth medium [176]. Electrical harvesting methods offer an attractive alternative to time-consuming and energy-intensive harvesting processes such as centrifugation and filtration [10]. However, additional research is required to make this technique economically viable for large-scale microalgae cultivation.

4.1.1.6. Comparison of physical methods based on various factors. Microalgae harvesting methods can be evaluated in terms of factors like energy usage, cost-effectiveness and efficiency; physical methods like centrifugation or flocculation tend to be popular due to their low energy use and efficient operation, as shown in Table 8.

Table 8
Comparison of various physical methods for microalgae harvesting [162,177,178].

Factors	Centrifugation	Filtration	Gravity sedimentation	Flotation	Electrical
	Rate				
Efficiency	High	Moderate	Low	Moderate to high	Moderate to high
Energy consumption	High	Low	Very Low	Moderate	Moderate to high
Capital cost	High	Low to moderate	Very Low	Moderate	High
Labor requirement	High	Low to moderate	Low	Moderate	Low to moderate
Scalability	Limited	High	Limited	Limited to high	Limited to high
Suitability for high biomass concentration	High	Low to moderate	Moderate to high	Moderate	Moderate to high
Suitability for shear sensitive microalgae	Low	High	High	High	Moderate
Suitability for harvesting large microalgae	Moderate to high	High	Moderate to high	High	Moderate to high

Table 9
Comparison of various Chemical methods, Biological method and Magnetic method for microalgae harvesting [11,103,118,120,178,182].

Factor	Chemical methods	Biological method	Magnetic method
	Rate		
Efficiency	High	Moderate to high	High
Energy consumption	Moderate to high	Low to moderate	Moderate
Capital cost	Moderate to high	Low to moderate	Moderate
Labor requirement	Low to moderate	High	Low
Scalability	Moderate to high	Low to moderate	High
Suitability for high biomass concentration	High	Low	High
Suitability for shear sensitive microalgae	Low	High	Moderate
Suitability for harvesting large microalgae	High	Low	High

4.2. Chemical methods

Chemical methods of microalgae harvesting involve the use of chemicals to flocculate or coagulate microalgae cells into aggregates or flocs that can easily be separated from the water. Such agents include metal salts (aluminum or iron), and polymers (chitosan or polyacrylamide) [179]. Chemicals are added to a culture medium and mixed thoroughly for even distribution. Then, the microalgae cells attach and either sink to the bottom or rise to the surface, depending on how dense the floc is [120]. Any biomass that has been harvested using sedimentation or flotation methods can be harvested again using either of these methods. Chemical methods may be fast and cost-effective solutions, but they require intensive monitoring to ensure that they do not negatively affect microalgae or the environment [11,120]. Furthermore, residual chemicals must be removed from harvested biomass before it can be used further for processing or as feedstock [8,180,181]. Microalgae harvesting by chemical methods: factors to consider, as shown in Table 9.

4.3. Biological methods

Biological methods of harvesting microalgae involve using natural compounds or organisms that selectively attach microalgae, making them easier to separate from the culture medium [183,184]. Some common ways to harvest microalgae biologically are:

- Flocculation: Flocculation is the process of adding flocculant agents like chitosan or organic polymers that cause microalgae to form larger groups that settle more easily than before [9,10].
- Bio-flocculation: This technique involves using microorganisms such as bacteria or yeast that produce extracellular polymeric substances (EPS), which induce microalgae to aggregate [185].

Biological methods of microalgae harvesting offer several advantages over chemical or mechanical approaches, including reduced environmental impact and cost-effectiveness [156]. Their effectiveness may depend on several factors, including the microalgae species type, biomass concentration level, and properties of the harvesting agent or organism used.

4.4. Magnetic methods

Magnetic methods of microalgae harvesting typically employ magnetic nanoparticles coated with materials that bind specifically to microalgae surfaces and enable easy separation by use of a magnetic field [120,186,187]. The procedure involves adding magnetic nanoparticles to microalgae cultures, followed by applying a magnetic field [188]. These nanoparticles bind with microalgae cells, while the magnetic field serves to pull them away from their culture medium. Magnetic methods of microalgae harvesting offer numerous advantages over traditional harvesting methods, including increased efficiency, lower energy usage, and less damage to microalgae cells [189]. Furthermore, magnetic nanoparticles can easily be recovered and reused to minimize waste and costs [190].

Overall, finding emphasize Microalgae harvesting is a vital step in the production of biofuels, food supplements, and medicines. Efficient and cost-effective harvesting techniques are crucial for the overall economics of production systems. Physical methods like centrifugation and filtration are commonly used but can be energy-intensive and costly. Biological methods offer eco-friendly alternatives, while magnetic methods show promise with their efficiency and reusability. The choice of harvesting method depends on factors such as microalgae species, culture conditions, and intended applications. Continuous improvement and optimization of these techniques will further enhance the economic viability and sustainability of microalgae-based industries. Research in this area remains important to develop even more efficient and eco-friendly harvesting solutions.

5. Microalgae cell disruption

Microalgae cell disruption involves breaking open microalgae cells to gain access to their intracellular components, including lipids, proteins, and carbohydrates [13–15]. A wide range of industries use these components, including biofuel production and distribution, pharmaceutical development, and cosmetic manufacturing. There are various methods for disrupting microalgae cells: mechanical, chemical, and biological methods [16,17].

5.1. Mechanical method

Microalgae cell disruption is an integral step in extracting their contents, including lipids, pigments, and proteins. In this method,

physical force is used to break the cell wall of microalgae to release the intercellular component [17,191,192]. Below are several mechanical methods for microalgal cell disruption.

- Bead milling: Bead beating is an established mechanical method used for cell disruption. Cells are exposed to mechanical agitation using glass or ceramic beads that generate mechanical shear forces that disrupt their integrity and cause disruption [193].
- High-Pressure Homogenization: High-pressure homogenization involves forcing microalgae cells through a narrow gap at high pressure, creating high shear forces that cause them to rupture and fragment. This technique generates shear forces that disrupt cells [194].
- Ultrasonication: Ultrasonication uses high-frequency sound waves to disrupt cells. The sound waves generate cavitation bubbles that collapse, releasing energy that disrupts their structure [195,196].
- Grinding: Ball milling is a technique used to disrupt microalgae cells using high-energy balls. Cells are loaded into a ball mill that rotates at high speed, and its balls collide with them and disrupt them [197].
- Microfluidization: Microfluidization is a technique that utilizes high-pressure fluid streams to disrupt cells. Under high pressure, they are forced through narrow gaps at high speed to disrupt them [198].
- Freeze-thaw Cycling: Freeze-thaw cycling exposes microalgae cells to repeated cycles of freezing and thawing, disrupting their structure by expanding and contracting water within their cells as the temperature fluctuations change [199].

5.2. Chemical methods

Chemical methods of microalgae cell disruption involve using chemicals to break down cell walls and release intracellular compounds [16]. Below are several chemical methods for microalgal cell disruption.

- Acids: Treatment with acidic substances such as sulfuric, hydrochloric, and nitric acids is one way to disrupt microalgae cells and release their contents [200,201]. These acids hydrolyze cell walls to release these components of microalgae cells while also denaturing proteins and decreasing yields. However, overuse of acids could result in protein denaturation and lower yields.
- Bases: Treatment with bases such as sodium hydroxide or potassium hydroxide may also help disrupt microalgae cells, as these agents break down their cell walls via saponification to release intracellular compounds [202,203]. However, prolonged use may result in protein denaturation.
- Organic Solvents: Organic solvents like ethanol, methanol, and acetone can be effective tools for disrupting microalgae cells by dissolving their cell wall lipids to release intracellular compounds [204]. However, their use could lead to denaturing proteins and decreased yields.

5.3. Biological method

Biological methods of microalgae cell disruption involve the use of enzymes [205]. Enzymes are biological catalysts capable of breaking down microalgae's cell walls. Cellulase, for instance, can attack and degrade cellulose found in some microalgae cell walls while other enzymes, like chitinase and lysozyme, can attack and break down chitin and peptidoglycan in other cell walls [206–208]. Enzymes offer several advantages when used to disrupt microalgae cells. Not only are they generally safe and environmentally friendly, but they can also be produced using renewable resources [16]. However, the effectiveness of enzyme-based cell disruption methods depends on both the type of microalgae present and which specific enzymes are utilized.

5.4. Comparative evaluation of microalgal cell disruption techniques: Advantages and disadvantages

Microalgal cell disruption is a key step in extracting intracellular components from microalgae cells, and evaluating various techniques can assist in choosing the one with the best advantages and disadvantages for each application, as shown in Table 10

Overall, finding emphasize Microalgae cell disruption and lipid extraction are crucial steps in obtaining valuable intracellular components like lipids, proteins, and carbohydrates. Various mechanical, chemical, and biological methods are used for cell disruption, each with its advantages and disadvantages. The choice of technique depends on the specific application and microalgae species. After disruption, lipid extraction methods like Folch, Bligh and Dyer, Soxhlet, supercritical fluid extraction, microwave-assisted extraction, and ultrasonic-assisted extraction are employed. Optimizing these processes is essential for sustainable and efficient production of valuable products from microalgae.

6. Lipid extraction

Lipid extraction from microalgae involves extracting oils or fats from microalgae cells for further processing [18,19]. Microalgae are an attractive source of lipids for use in biofuel production and other high-value applications, such as cosmetics. Lipid extraction involves multiple steps, including solvent extraction and purification. Lipids are separated from cell debris using solvent extraction [210,211]. Organic solvents like hexane, chloroform, and methanol are frequently employed for this purpose [212]. Microalgal biomass is combined with a solvent to dissolve its lipids, and then the resultant mixture is filtered to remove any remaining cell debris [213]. The extracted lipids are then purified through processes such as chromatography, distillation, or evaporation to obtain a pure lipid product [214]. The purity of the final product depends on the quality of the starting material and the effectiveness of the extraction and purification steps.

6.1. Extraction methods for total lipid

Extraction methods for total lipids are techniques used to isolate all forms of lipids found in biological samples, such as fatty acids, triglycerides, phospholipids, and cholesterol. Some methods of extraction that are commonly used.

- Folch Extraction: This technique involves dipping samples into methanol and chloroform to dissolve lipids and then separating them. The Folch method's performance is based on the presence of mineral salts in the crude extract and the use of a significant amount of solvent. Therefore, this method is widely employed for extracting lipids from biological samples [215].
- Bligh and Dyer extraction: An alternative organic solvent extraction method is needed for larger-scale use because Bligh and Dyer's method, traditionally used to extract lipids from biological matrices, generates significant quantities of waste solvent. Although effective for microalgal lipid extractions, this method is costly to recycle and can be unsafe to handle in large amounts. Similar to Folch's extraction technique but requiring less solvent, the proposed method extracts lipids from materials using chloroform, methanol, and water instead of chloroform, chloromethanol, and water. Bligh and Dyer's method has been used as a benchmark for comparing solvent extraction methods [216,217].
- Soxhlet extraction: The Soxhlet extraction method, which involves the continuous extraction of non-polar lipids using a heated solvent, is one of the chemical methods used for lipid extraction. Various chemical solvents, such as hexane and petroleum ether, are used in this method to extract oil. In the process, dried

Table 10
The advantages and disadvantages of different types of microalgal cell disruption methods [16,191,209].

Method	Advantages	Disadvantages
Mechanical cell disruption	• High yield • Scalable • Can be optimized • No chemical residues	• Energy-intensive • Can damage sensitive products • May require specialized equipment
Chemical cell disruption	• High efficiency • Low energy consumption • Low energy consumption • Selective extraction	• Can damage sensitive products • May require hazardous chemicals • Chemical residues in the product
Biological cell disruption	• Mild conditions • No toxic chemicals • High selectivity • Low energy consumption • Can be integrated with cultivation	• Low yield • Limited scalability • Can be slow

algae are placed in an algae reservoir and hexane is added, causing the algae to completely immerse themselves in the solvent. As only a small amount of the algae dissolves in the hexane, the hexane is boiled to vaporize it, while the oil, which has a higher boiling point, is extracted through the Soxhlet extraction process [218,219].

- Supercritical fluid extraction: supercritical fluid extraction is the process of separating one component from another using a supercritical fluid. The most commonly used supercritical fluid is carbon dioxide, which can be modified with ethanol or methanol. This solvent has several advantages, including non-toxicity, non-explosiveness, non-flammability, low cost, availability, and ease of removal from extracted oils. The basic principle of supercritical fluid extraction is that when the feed material comes into contact with the supercritical fluid, volatile substances partition into the supercritical phase. After the soluble material is dissolved, the supercritical fluid is removed from the feed material using pressure and temperature changes. This relatively new method extracts lipids from samples using supercritical fluids, such as carbon dioxide, and offers benefits over conventional extraction methods, including lower dependence on organic solvents and shorter extraction periods [220].
- Microwave-assisted extraction: Microwave extraction, also known as microwave-assisted extraction (MAE), is a novel extraction method that combines traditional solvent extraction with microwave technology. It has gained popularity in lipid extraction research due to its ability to rapidly and efficiently heat and remove lipids from samples using microwave energy [221].
- Ultrasonic-assisted extraction: This method utilizes ultrasonic waves generated by ultrasonic reactors to extract oil. First, ether is added to dried algae and subjected to ultrasonic waves for 30 min. The ultrasonic waves create cavitation bubbles in the solvent material, causing the cell walls of the algae to rupture. Afterward, the mixture is filtered with sieves, and the algae are manually pressed to extract the algal oil. This oil can then be used in further transesterification processes. The technique is effective for disrupting microalgal cells and extracting bioactive compounds. It uses high-frequency sound waves, ranging from 20 kHz to 100 kHz, to break down the cell walls of the algae and release their contents into a solvent [222,223].

Overall, Microalgae cell disruption is vital for accessing valuable intracellular components used in biofuel, pharmaceuticals, and cosmetics. Mechanical, chemical, and biological methods are employed for this purpose. Lipid extraction from microalgae involves solvent extraction and purification using various techniques. Optimizing these processes is crucial for maximizing microalgae's potential in different industries.

7. Transesterification

Transesterification is an essential process used in biodiesel production. It involves the conversion of triglycerides derived from plant or animal fats to fatty acid methyl esters (FAMEs), the main constituents of biodiesel [20–22]. The reaction can be catalyzed using acid or base catalysts, typically such as sodium hydroxide (NaOH) or potassium hydroxide (KOH) [25,224]. The chemical equation for the transesterification reaction is given [225,226]:

Triglyceride + Alcohol → Fatty Acid Esters + Glycerol

$C_3H_5(CO_2R)_3 + 3\ CH_3OH \rightarrow 3\ CH_3(CO_2R) + C_3H_5(OH)_3$

Where R is a long-chain alkyl group, typically a combination of C_{16} and C_{18} chains.

In this equation, triglyceride ($C_3H_5(CO_2R)_3$) reacts with three molecules of methanol (CH_3OH) to form three molecules of methyl ester (biodiesel), with one remaining as glycerol ($C3H_5(OH)_3$) as a byproduct. Reactions typically take place between 50–60 °C for several hours, using catalysts such as sodium hydroxide (NaOH) or potassium hydroxide (KOH) to accelerate the process and help produce desired products [227,228]. Alcohols used in this reaction process typically methanol or ethanol act as nucleophiles attacking carbonyl groups in triglycerides and forming ester bonds between the fatty acid ester. This results in the formation of biodiesel esters, with byproducts such as glycerol remaining.

Overall, Transesterification is crucial in biodiesel production, converting triglycerides to fatty acid methyl esters (FAMEs). Acid or base catalysts like sodium hydroxide or potassium hydroxide accelerate the reaction with alcohols (usually methanol or ethanol) to form biodiesel and glycerol. This process plays a vital role in sustainable energy solutions, utilizing renewable feedstocks for biodiesel production.

8. Challenges, future opportunities

Algal biodiesel has long been recognized as a viable alternative to fossil fuels due to its high productivity, low land-use requirements, and adaptability to various environmental conditions [229]. However, developing and commercializing algae biodiesel presents both challenges and opportunities in its production and commercialization processes.

Challenges

- High Production Costs: Algae biodiesel production currently has less economic viability due to its much higher production costs compared to traditional fossil fuels [230–232].
- Limited Scalability: Even though algae production can be increased on a small scale, it is hard to meet the needs of the global fuel market right now [233].
- Energy-Intensive Harvesting: Energy resources of large magnitude are needed for the harvesting of algae for oil production [10].
- Efficiency: The efficiency of the production process is another issue with biodiesel. Its production may become expensive due to the time and resources required for cultivation [155].
- Water Usage: Algae need a lot of water to grow, which can be problematic in places with scarce water resources [234,235].

- Contamination risks: Algae are extremely susceptible to contamination by other species, which lowers the yield and quality of produced biodiesel [236].
- Technical Challenges: Ongoing development of efficient and cost-effective algae cultivation and harvesting technologies includes the creation of systems that can withstand severe environmental conditions and the optimization of growth conditions to increase oil production [189,237].
- Infrastructure: production, transport, and usage of algae biodiesel need significant investments before they can be properly implemented [238].

Future Opportunities

- Carbon Neutrality: As it takes in carbon dioxide from the environment while growing, algae biodiesel has the potential to be a carbon-neutral fuel source [239,240].
- Renewable and Sustainable: Algae is an abundant source of renewable energy that could reduce our reliance on nonrenewable fossil fuels [241].
- Versatile Applications: Biodiesel derived from algae has multiple applications, including everything from transportation as well as heating to electricity generation [85,242].
- Effective Use of Land: The cultivation of algae can make use of land that would otherwise go unused by allowing them to grow where conventional crops cannot [41,243].
- Research and development: Biofuel could become a more affordable fuel alternative with continued research and development into the growth of algae and the manufacturing of biodiesel [116].
- Co-Products: In addition to producing biofuel, algae may be used for other purposes. Algae, for instance, could be used as fertilizer or even as a source of protein for animal feed [244].
- Waste Utilization: Waste materials from other industries, such as wastewater and carbon dioxide, can be used in the manufacturing of algae biodiesel, offering a more economical and environmentally friendly alternative [244].

8.1. Application of findings

The findings of this study offer significant implications for the application of algal biodiesel in practical scenarios. By addressing the challenges identified, such as high production costs, limited scalability, and energy-intensive harvesting, we can pave the way for more economically viable and sustainable production processes. For instance, the optimization of algae cultivation techniques and the development of cost-effective harvesting technologies will be crucial in enhancing the overall efficiency of biodiesel production. These measures can potentially reduce the economic barriers and make algal biodiesel a competitive alternative to conventional fossil fuels. Moreover, the study's emphasis on algae's carbon neutrality and renewable nature highlights the environmental benefits of this biofuel. Policymakers and industries can capitalize on this advantage to accelerate the transition towards cleaner energy sources and reduce greenhouse gas emissions. The versatile applications of biodiesel derived from algae, including transportation, heating, and electricity generation, underscore the potential to diversify energy use and achieve greater energy security. Additionally, the research and development efforts highlighted in the study underscore the importance of continued investments and innovation in this field. With ongoing advancements, algal biodiesel could become a more affordable and widely adopted fuel alternative in the future. The co-products potential of algae, such as its use as fertilizer or a source of protein for animal feed, further demonstrates the multifaceted benefits that can be derived from algal biomass, promoting a circular and sustainable economy. Furthermore, the utilization of waste materials, like wastewater and carbon dioxide, in algae biodiesel production offers a promising approach to minimize environmental impact and create a more circular and resource-efficient system. This eco-friendly strategy can contribute to waste management solutions and support the growth of algae-based biofuel industries while mitigating their overall environmental footprint. Overall, the application of the findings from this study holds immense promise for the practical integration of algal biodiesel into our energy landscape. By addressing the challenges and capitalizing on the opportunities presented, we can foster a greener, more sustainable energy future that relies on renewable resources and reduces our dependency on non-renewable fossil fuels. The potential benefits extend beyond energy production, encompassing environmental conservation, waste utilization, and sustainable agriculture, making algal biodiesel a pivotal component of a holistic approach towards a cleaner and more resilient future.

Overall, the findings emphasize the challenges and opportunities in algal biodiesel production. To make it economically viable, optimization of cultivation and harvesting techniques is crucial. Algal biodiesel's carbon neutrality and versatility offer potential for various applications, from transportation to agriculture. Continued research and development can lead to cost reductions and wider adoption. Moreover, utilizing waste materials in production promotes a circular and sustainable approach. Overall, algal biodiesel holds promise as a cleaner and more resilient energy alternative.

Concluding remarks

The production of biodiesel from microalgae has shown great promise as an eco-friendly and sustainable alternative to fossil fuels, but it faces some challenges associated with its commercialization. There are many challenges associated with the production of biodiesel from microalgae, such as high production costs, low productivity, strain selection issues, and cultivation techniques that need to be improved. Even so, the potential benefits of reducing greenhouse gas emissions, increasing energy security, and enhancing rural development cannot be underestimated. The use of microalgae biodiesel may eventually become part of our energy mix if research and development efforts are successful. This emerging technology should be prioritized by policymakers and investors to enable its successful deployment over the next few years.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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